

Project 1 — Technical Report

Dex3 to BrainCo Revo2 Touch Embodiment Transfer

Assignment: Unitree G1_29DoF Take-Home Assessment — Assignment 1

Repository: `/home/autonomique/jp_test`

Final Evaluation Report — Project 1 (Guide §10)

Date: 2026-08-31

Workspace: `/home/autonomique/jp_test`

Executive summary

Item	Result
Final eval pass	yes
Stage A (Dex3 source)	100% success, 4 configs × 10 trials
Stage B (BrainCo retarget)	Playback pass, 0 limit violations
Stage C (fine-tuning)	Not run (optional per guide §9)
All phases 0–7 verified	yes (after checkpoint field fixes)

Guide §10 — A/B/C results table

See [results/final/project1_abc_results.md](#) for the full table.

Metric	A: Dex3	B: BrainCo retarget	C: + FT
Success rate	100%	100% (playback)	not run
Object drops	0	N/A	—
Joint-limit violations	0	0	—

Metric	A: Dex3	B: BrainCo retarget	C: + FT
Retarget smoothness	N/A	2.4 cm max Δ /frame	—

Stage A configurations (§4.3 + §10)

Config	Success	Drops	Mean lift
C1 nominal	100%	0	6.0 cm
C2 pose variation	100%	0	6.0 cm
C3 mass variation	100%	0	6.0 cm
C4 held-out pose	100%	0	6.0 cm

Artifacts: `results/stage_A/stage_a_baseline.json`,
`checkpoints/stage_A/demos/.../data.json`

Per-phase verification

Phase	Guide section	Status	Key artifact
0	Stack freeze	pass	<code>results/phase0/stack_verification.json</code>
1	G1+Dex3 bring-up	pass	<code>results/phase1/dex3_sanity_test.json</code>
2	Stage A baseline	pass	<code>results/stage_A/stage_a_baseline.json</code>
3	BrainCo bring-up	pass	<code>results/phase3/brainco*_hand_test.json</code>
4	Correspondence	pass	<code>configs/correspondence_table.yaml</code>
5	Retargeter	pass	<code>results/phase5/retarget_validation.json</code>
6	Stage B playback	pass	<code>results/stage_B/stage_b_baseline.json</code>
7	G1+BrainCo coupling	pass	<code>results/phase7/g1_brainco_coupling.json</code>
8	Final eval	pass	<code>results/final/project1_abc_results.json</code>

Run verification:

```
python3 scripts/final/verify_all_phases.py
```

Failure analysis (Stage B taxonomy, §8)

Code	Mode	Mitigation
F1	Morphology mismatch	Semantic + task-space retargeting
F4	Object slip	Hybrid palm-attached / scripted lift (Stage A)
F7	Arm-hand reachability	Phase 7 wrist coupling; full USD swap deferred

Known limitations (explicit)

1. Stage A uses kinematic lift assist — not pure contact grasp.
2. Stage B evaluated as BrainCo hand trajectory playback (standalone USD); full G1+BrainCo pick-place on one articulation requires vendor USD swap.
3. Stage C (residual adapter fine-tuning) not implemented — guide allows A/B-first delivery.
4. Task-space IK v2 (`task_space_ik_v2`) implemented but requires separate Isaac sessions per hand; production path uses `semantic_joint_map_v1` (validated).

Reproduce final evaluation

```
cd /home/autonomique/jp_test
bash scripts/final/run_final_eval.sh --headless --device cuda --enable_c
```

Outputs land in `results/final/`.

System Architecture — Project 1

Data flow

```
flowchart LR
    subgraph StageA["Stage A — Dex3 source"]
        DEMO["Stage A demo\n(data.json)"]
        EVALA["evaluate_stage_a.py\nC1-C4 configs"]
    end

    subgraph Retarget["Retargeting layer"]
        CACHE["Dex3 tip cache"]
    end
```

```

    JMAP["semantic_joint_map_v1"]
    IK["task_space_ik_v2"]
    ADAPT["residual_adapter_v1\n(Stage C)"]
end

subgraph StageB["Stage B – BrainCo retarget"]
    DEMOB["Stage B demo"]
    PLAY["BrainCo hand playback"]
end

subgraph StageC["Stage C – Fine-tuned"]
    DEMOC["Stage C demo"]
end

subgraph Sim["Isaac Sim / Isaac Lab"]
    G1["G1 + Dex3 env"]
    BC["BrainCo standalone USD"]
    COUP["G1 wrist coupling\n(Phase 7)"]
end

DEMO --> CACHE
DEMO --> JMAP
CACHE --> IK
JMAP --> DEMOB
IK --> DEMOB
JMAP --> ADAPT
ADAPT --> DEMOC
DEMO --> EVALA
G1 --> EVALA
DEMOB --> PLAY
BC --> PLAY
DEMOB --> PLAY
DEMO --> G1
DEMOB --> COUP

```

Component responsibilities

Component	Role	Frozen / trainable
Source demo	Motion prior (xr_teleoperate format)	Frozen
Dex3 scripted policy	Stage A trajectory generator	Frozen

Component	Role	Frozen / trainable
Joint-map retargeter	Dex3 7-DoF → BrainCo 6-DoF	Frozen (deterministic)
Task-space IK	Tip-target optimization	Frozen (deterministic)
Residual adapter	$q^* = q_{\text{joint}} + f(q_{\text{joint}})$	Trainable (Stage C)
BrainCo sim playback	Stage B/C validation	Eval only

Evaluation loop

```
configs/stage_a_eval.yaml → Stage A trials → results/stage_A/  
configs/finetune.yaml → Stage C train → checkpoints/stage_C/  
retarget + validate → Stage B demo → results/stage_B/  
scripts/final/evaluate_final_abc.py → results/final/project1_abc_resul
```

Key paths

- Source: checkpoints/stage_A/demos/.../data.json
- Retarget: checkpoints/stage_B/demos/.../data.json
- Fine-tuned: checkpoints/stage_C/demos/.../data.json
- Checkpoints: checkpoints/stage_C/
residual_adapter_{right,left}.pt

Sim-to-Real Deployment Plan — G1 + BrainCo Revo2 Touch

1. Calibration

Item	Procedure
BrainCo joint zeros	Match URDF zero to open-hand pose; verify against configs/g1_brainco_target.yaml limits
G1-hand mount	Use Phase 7 wrist poses (results/phase7/ g1_brainco_coupling.json) as SE(3) mount transform

Item	Procedure
Dex3→BrainCo scale	Re-run correspondence verification (<code>scripts/phase4/verify_correspondence.py</code>) on hardware

2. Control stack

- **Rate:** 100 Hz joint targets (per `configs/source_policy.yaml`)
- **Mode:** Position control on BrainCo actuated joints (6 DoF per hand)
- **Retargeting:** Offline demo replay first; on-robot: `q_cmd = q_retarg + adapter(q_retarg)` at 100 Hz

3. Latency budget

Stage	Budget
State read	≤ 5 ms
Retarget + adapter	≤ 3 ms (pre-computed table or small MLP)
Command write	≤ 2 ms
Total	≤ 10 ms (100 Hz)

4. Safety limits

- Clamp all commands to `g1_brainco_target.yaml` joint limits
- Velocity cap: 50% of URDF velocity limits during bring-up
- E-stop on: joint limit violation, fall detection (IMU), operator override
- Disable adapter until open/close sanity passes

5. Staged hardware validation

1. **Week 1:** BrainCo open/close, per-joint sign check (mirror Phase 3 scripts on real hand)
2. **Week 2:** Fixed-base G1 arm replay; BrainCo mounted; no object
3. **Week 3:** Reach + pre-grasp near object; no lift
4. **Week 4:** Full pick-place with human supervision; log drops and retarget errors

6. Known sim gaps before hardware

- Kinematic lift assist in Stage A (not contact-based grasp)

- Full G1+BrainCo single-articulation USD not in public assets
- Stage B/C validated as hand trajectory playback; whole-body contact not re-validated on BrainCo body

7. Data collection for adapter refinement

- Collect 5–10 teleop demos on real BrainCo after mount
- Fine-tune residual adapter only (keep retargeter frozen)
- Compare fingertip/proprio error vs sim Stage C baseline

Assessment Compliance — Project 1 vs Technical Assessment PDF

Assignment: Dex-hand → BrainCo Revo2 Touch embodiment transfer (Assignment 1)

Deadline: August 31, 2026

Can Project 1 be finished?

Yes for Stages A, B, and C, with documented deviations on task choice and full USD swap.

Requirement (PDF)	Status	Evidence
§1.2 Named source policy/demo	■	configs/source_policy.yaml , results/phase2/source_audit.json
§1.3 Retarget without fine-tuning (Stage B)	■	results/stage_B/stage_b_baseline.json
§1.4 Evaluate across ≥3 configs + held-out	■	C1-C4 in results/stage_A/stage_a_baseline.json
§1.5 Fine-tune retargeted policy (Stage C)	▲	results/stage_C/stage_c_baseline.json , checkpoints/stage_C/residual_adapter_*.pt
	Partial	

Requirement (PDF)	Status	Evidence
§1.1 Full G1 USD hand swap	■	Standalone BrainCo + wrist coupling (results/phase7/)
Example: bimanual handover	Deviation ■	Pick-place cylinder task used instead (documented)
M1–M6 metrics	A/B/C	results/final/project1_abc_results.md
Videos per config	■ ■	results/videos/video_manifest.json — MP4 (Stage A C1–C4 + B/C BrainCo camera)
Architecture diagram	■	docs/architecture.md
Technical report		docs/technical_report.md , docs/technical_report.pdf
Sim-to-real plan		docs/sim_to_real_plan.md

Reproduce core pipeline

```
bash scripts/final/run_final_eval.sh --headless --device cuda --enable_c
python3 scripts/final/verify_all_phases.py
```

Known code limitations (explicit)

1. Stage A uses kinematic lift assist — not pure contact grasp.
2. Stage B = BrainCo hand trajectory playback (not full G1+BrainCo pick-place on one articulation).
3. Stage C residual adapter fine-tunes joint trajectories (570-frame teacher from IK v2); not full RL in sim.

Priority 1 & 2 deliverables

```
bash scripts/submission/run_priority1.sh --headless --device cuda --enab
bash scripts/phase8/run_phase8.sh --headless --device cuda --enable_came
```


Project 1 — Final A/B/C Results

(Guide §10)

Final eval pass: yes Generated: 2026-08-31T11:38:10.572442+00:00

A/B/C comparison table

Metric	A: Dex3 source	B: BrainCo retarget	C: BrainCo + FT
Task / playback success rate	100.0%	100.0%	100.0%
Fingertip tracking error (proxy, m)	N/A (source FK not aggregated)	1.20 cm (joint playback)	1.24 cm (joint playback)
Wrist tracking error (m)	N/A	0.59 cm (mount samples)	see phase7 (unchanged)
Object drops (mean per config)	0	N/A (playback-only Stage B)	N/A (playback-only)
Joint-limit violations (retarget)	0	0	0
Max frame delta (retarget smoothness, m)	N/A	1.83 cm	1.83 cm (inherited)
Mean completion time (s)	30.5	4.8	43.3 eval
Adaptation samples / time	N/A	mapping only	570 frames, 141.2s train
Retarget method	xr_teleoperate-format scripted demo	task_space_ik_v2	residual_adapter_v1

Stage A per configuration

- **C1_nominal**: success=100%, drops=0, lift=6.0cm
- **C2_pose_variation**: success=100%, drops=0, lift=6.0cm
- **C3_mass_variation**: success=100%, drops=0, lift=6.0cm
- **C4_held_out_pose**: success=100%, drops=0, lift=6.0cm

Failure analysis

- F1 morphology: BrainCo 6-DoF vs Dex3 7-DoF — mitigated by task-space retargeting
- F4 object slip: Stage A uses kinematic lift assist (not pure contact grasp)
- F7 reachability: G1+BrainCo full USD swap deferred; Phase 7 validates wrist coupling